



Project 01: Toca a Tiempo

Typology: Industrial Product Design

Course: Design Methodology (September–December 2025)

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TEXTOS:

The Challenge

Following field visits and direct research in adult day care centers, it was identified that a large portion of users experience advanced physical and cognitive decline (such as hearing loss, visual impairment, osteoarthritis, and tremors). Current technological leisure options often generate frustration and dependency, as they require complex setups, WiFi networks, or mobile applications controlled by third parties. Similarly, the market frequently offers products with childish aesthetics that devalue the dignity of elderly users.

Project Description

Toca a Tiempo was born as a hybrid between traditional cognitive games and electronic interactivity. It consists of a completely autonomous physical device that eliminates the need for smartphones or secondary screens. Its aesthetics steer away from childish tropes to draw inspiration from the clean interface of a classic TV game show or a user-friendly control panel, completely eliminating the "fear of breaking it" and opening an accessible door to cognitive stimulation.

Gameplay Mechanics and Interactions

The board's operation is intuitive and provides immediate sensory feedback:

- **Memorable Stimulation:** The board features a matrix of 21 translucent buttons that light up in blue, configuring a random pattern. The user has 30 seconds to memorize it.
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- **Immediate Feedback:** Once the lights turn off, the user must replicate the sequence. If correct, the buttons turn green; if they fail, the panel blinks red, granting new opportunities without stressful penalties.
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- **Accessibility at the User's Pace:** On the left side, 3 functional buttons allow the user to recall the remaining sequence for 5, 10, or 15 seconds. A touch sensor on the right side generates a new game instantly.

Ergonomy and materials selection

Every component has been meticulously selected through an ergonomic and anthropometric study to adapt to the actual capabilities of the target user:

- **ABS Plastic Casing:** Manufactured via injection molding, it offers a rigid, lightweight, and impact-resistant outer shell with a textured non-slip finish.
- **Elastomeric Silicone Buttons:** A matrix featuring large-surface buttons (42×42 mm) with a soft touch that requires minimal actuation force, making them ideal for hands with osteoarthritis or tremors.
- **Neutral Posture:** Optimized dimensions (550 × 240 × 60 mm) are designed so that, while the user is seated, the entire board remains within their frontal reach without needing to stretch, effectively preventing lumbar fatigue.
- **Visual Comfort:** Internal LEDs emit a homogeneous and diffuse light, ensuring sharp contrast for people with visual deficits or cataracts without causing annoying glare.

Conclusions

In short, **Toca a Tiempo** demonstrates how industrial design can be transformed into a tool for social impact with the capacity to improve people's quality of life. By balancing fully adapted ergonomics, an intuitive electronic interface, and solid commercial viability, the project transcends the concept of a simple interactive product. Its true value lies in democratizing access to cognitive stimulation, eliminating the barriers of the digital divide, and restoring autonomy, enjoyment, and dignity to the leisure time of older adults.